Edge-AI Motor Monitoring

Brief summary of what the project code does now

1. Read vibration

The ESP32 reads the MPU6050 accelerometer on GPIO 21 (SDA) and GPIO 22 (SCL). It samples the X axis at 1 kHz in 1,024-sample windows and removes the static offset.

2. Extract features and classify

Firmware computes 12 time and frequency features, then runs the compact Extra Trees model embedded in **sketch/model.h**. It can report normal, imbalance, misalignment, bearing outer-race, or bearing-ball conditions.

3. Detect stopped motor

A separate vibration-floor rule reports **motor_stopped** when RMS is below 0.015 g. Motor-off readings observed on this board were about 0.0038 g RMS. This state is not treated as a fault and keeps the buzzer off.

4. Alert and send status

Fault classifications light GPIO 27 and pulse the two-pin buzzer on GPIO 25. Normal operation lights GPIO 26. The ESP32 publishes the class, confidence, and alert flag to MQTT topic **esp32motor_7f3a/status**.

5. Show the dashboard

The Streamlit dashboard listens to MQTT and displays condition, confidence, alert state, and recent telemetry. It also has a simulator mode.

Current board check

Board / port	ESP32-D0WD-V3 on COM3
Firmware upload	Build succeeded; upload verified
Motor-off result	Serial reported motor_stopped, confidence 1.00, alert false
Dashboard	Running at http://127.0.0.1:8501

Model performance and limits

The latest recording-held-out evaluation scored 82.5% accuracy and 0.820 macro-F1 on 40 windows from five held-out recordings. This is a small dataset check, not a guarantee of accuracy on the assembled motor. The stopped state uses an RMS threshold rather than a newly trained class; validate it again when the motor is running. This firmware monitors the motor and does not power or start it.

Main files: sketch/sketch.ino, sketch/edge_features.h, sketch/model.h, dashboard.py, train_hardware.py. Project folder: C:/Users/DELL/Downloads/iot project.